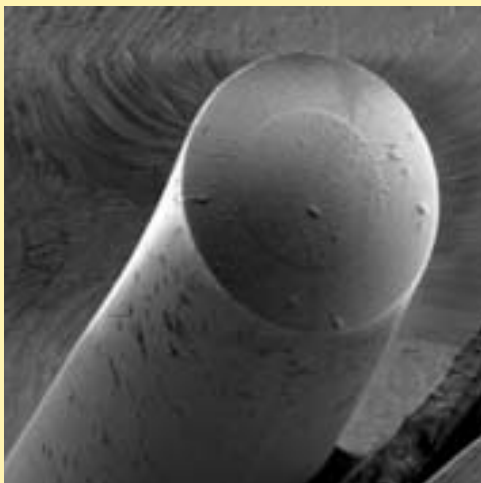


Optical pathways

The missing element in the discussion so far has been optical pathways that can interconnect different components, which are essential if a fully-optical computer is to be realised. As an example, consider the work being done by a group in Belgium, called the IO project (Interconnect by Optics). The group aims at developing 'an optical interconnect family' that will interconnect ICs to ICs, boards to boards, and more. The group aims at demonstrating direct access to a CMOS chip. Their approach is based on optical waveguides that can be integrated with boards.

The novelty here is in the optical waveguides. These will be based on plastic optical fibres—as opposed to traditional glass fibres—and on layered



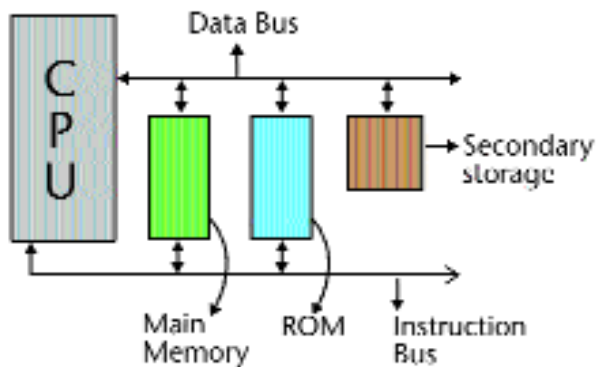
From the IO group, a highly magnified photograph of a plastic optical fibre (POF). The core and the cladding are visible

glass sheets with the waveguides etched on them.

The group focuses on the achievable channel density and data rate. The IO project aims to demonstrate a complete optical interconnect family in the area of core IP routers, which marks the entire project as not being something futuristic and 'in-theory', but one whose results can be used right now.

However, the point of this is not that we now have a faster technology in the area of core IP routers; the point is that research is paving the way for silicon-optical hybrid boards that might very well be available a few years from now.

The von Neumann architecture for computing machines



This figure seems simple and obvious, but it was not obvious until John von Neumann devised the architecture, known since 1945 as 'the von Neumann architecture'. Every computer you've ever seen is a 'von Neumann machine'. Parallel-processing machines are an exception

different research groups are different in their focus. Some groups focus on materials, some on interconnects, some on components, and so on.

The Internet grows exponentially, with users being added by the minute. With increasing demands on bandwidth, optical switching over fibre-optic networks will undoubtedly be one of the major focal points of research. However,

that is beside the point: when can we expect an optical computer?

Time frames vary. Holographic storage could be commercially available in as little as five years. Pure-optical processors have already been devised and built, and the way to a general-purpose optical processor has already been paved—although much work remains to be done—but that is not the issue here. What is at issue is the framework for an optical computer and its applications, which no single group is focusing on.

No one knows what an optical computer will be like, and what uses it will be put to. However, we may hazard a guess that it might not be based on the von Neumann architecture that we're all so familiar with. The primary reason for this is the speed that is promised by optical components, and the manner in which devices in such a computer can be interconnected.

As an example, with storage going the holographic way, will there be a distinction between main and peripheral memory? As of now, our concept of a computer is centred around a fast main memory and a slower hard disk. Optical storage might do away with that concept altogether.

Then there is the question of the network being totally distinct from the computer. With high-speed optical switching and optical processors, what might not be possible? The same processor that handles signals within the computer could also directly handle those coming in from and going out to the network, effectively uniting processor and network. The possibilities are exciting.

In sum, the picture we have of optical computing, as it stands today, is first, that the opticalisation of computers is happening component by component; there is a tremendous need for new materials that can be deemed as being capable of implementing what is being done in theory. And second: that we don't yet have a clear idea what a fully-optical computer will be like, but the very architecture that John von Neumann spelt out in 1945 may be challenged. ■

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